

ZEEBRUGGE, Belgium

WMO: 064180

Lat: 51.347N

Lon: 3.202E

Elev: 9

StdP: 101.22

Time Zone: 1.00 (EUC)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-3.6	-1.9	-7.7	2.0	-2.9	-5.4	2.4	-1.4	15.5	7.6	14.5	7.2	6.7	80	0.684

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.6	24.8	19.2	22.9	18.4	21.6	17.7	20.0	24.2	19.1	22.1	18.3	21.0	5.2	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.4	13.3	22.1	17.7	12.7	21.1	17.0	12.1	20.4	57.4	24.3	54.2	22.0	51.7	21.0	23.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.0	12.5	11.3	DB	-4.4	29.8	3.0	1.8	-6.6	31.1	-8.4	32.2	-10.1	33.2	-12.3	34.5	(4)
(5)			WB	-4.9	22.1	2.8	0.9	-7.0	22.7	-8.6	23.2	-10.2	23.8	-12.3	24.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.4	4.9	4.6	6.9	10.1	12.8	15.8	18.3	18.3	16.2	13.1	8.9	6.1
	DBStd	5.61	3.25	3.32	3.02	2.89	2.82	2.32	2.25	1.95	2.16	2.64	2.85	3.25
	HDD10.0	633	159	153	101	33	4	0	0	0	0	6	53	124
	HDD18.3	2612	418	386	353	248	174	83	27	24	72	164	284	380
	CDD10.0	1129	1	1	6	35	92	174	256	257	185	101	19	3
	CDD18.3	64	0	0	0	0	2	7	25	22	7	1	0	0
	CDH23.3	153	0	0	0	0	4	16	76	39	18	1	0	0
	CDH26.7	33	0	0	0	0	0	2	22	6	3	0	0	0
Wind	WSAvg	6.1	6.7	6.6	6.1	5.6	5.8	5.9	5.5	5.5	5.7	6.1	6.5	7.0
Precipitation	PrecAvg	817	66	59	48	43	57	67	75	81	69	77	90	84
	PrecMax	965	131	112	126	81	116	140	159	186	194	140	166	181
	PrecMin	641	2	16	10	2	19	25	19	9	10	19	10	20
	PrecStd	93	32	26	27	22	24	30	36	48	44	30	35	43
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.1	12.4	16.1	20.2	23.4	25.7	29.9	27.1	25.6	22.1	16.5	13.2
		MCWB	10.7	9.1	11.2	13.6	16.6	20.1	20.8	20.9	19.5	17.0	14.7	11.6
	2%	DB	10.8	10.8	13.7	17.8	20.0	21.9	24.9	24.1	22.4	19.5	15.0	12.3
		MCWB	9.7	8.8	10.3	12.7	15.6	17.9	19.6	19.1	18.2	16.0	13.3	10.9
	5%	DB	9.7	9.7	11.8	15.9	18.1	20.4	22.9	22.4	20.8	17.8	14.0	11.4
		MCWB	8.5	8.0	9.3	11.8	14.7	16.9	18.5	18.4	17.3	15.1	12.4	10.1
	10%	DB	8.9	8.7	10.6	13.9	16.7	19.0	21.4	21.2	19.3	16.6	12.9	10.3
		MCWB	7.7	7.2	8.6	10.9	13.9	15.8	17.6	17.5	16.2	14.5	11.3	9.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.0	10.2	11.8	14.5	17.6	20.3	21.4	21.1	20.3	17.6	14.9	12.1
		MCDB	11.8	11.3	15.4	19.1	20.4	24.7	27.1	25.8	24.6	21.2	15.9	12.9
	2%	WB	9.8	9.0	10.7	13.1	16.2	18.7	20.1	19.7	18.8	16.2	13.5	11.1
		MCDB	10.6	10.3	13.1	17.1	19.4	21.6	24.7	23.2	21.8	18.9	14.8	12.1
	5%	WB	8.7	8.1	9.7	12.1	15.1	17.2	19.0	18.9	17.7	15.4	12.3	10.2
		MCDB	9.5	9.6	11.4	15.2	17.9	19.6	22.1	21.6	20.1	17.4	13.7	11.3
	10%	WB	7.9	7.2	8.8	11.2	14.1	16.2	18.2	18.1	16.7	14.7	11.3	9.1
		MCDB	8.8	8.5	10.4	13.8	16.6	18.5	20.9	20.7	19.1	16.4	12.6	10.2
Mean Daily Temperature Range	5% DB	MDBR	3.2	3.8	4.2	4.8	4.3	4.1	4.6	4.9	4.7	4.1	3.5	3.3
		MCDBR	3.8	5.1	6.0	7.9	6.7	6.6	6.9	6.6	6.8	5.6	4.3	3.5
		MCWBR	3.9	4.1	4.1	4.6	3.8	4.1	4.0	3.8	4.1	3.8	3.9	3.4
	5% WB	MCDBR	3.9	4.9	5.5	6.8	5.8	5.7	6.1	5.9	5.7	5.1	4.0	3.5
		MCWBR	4.1	4.0	4.2	4.4	3.7	4.0	3.9	3.9	3.9	3.9	3.8	3.7
Clear-Sky Solar Irradiance	taub	0.335	0.354	0.390	0.416	0.425	0.421	0.424	0.415	0.393	0.371	0.347	0.330	
	taud	2.372	2.357	2.262	2.227	2.253	2.283	2.284	2.325	2.373	2.411	2.382	2.348	
	Ebn at Noon	671	762	797	821	831	836	827	814	789	733	647	608	
	Edh at Noon	64	83	109	126	129	126	124	114	97	78	63	56	
All-Sky Solar Radiation	RadAvg	0.76	1.46	2.70	4.42	5.18	5.60	5.29	4.52	3.32	1.89	0.89	0.57	
	RadStd	0.06	0.26	0.34	0.39	0.37	0.43	0.53	0.37	0.22	0.16	0.09	0.07	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (71 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page